

Dr. Doc:

Embryological Development of the Lower Respiratory Tract

The lower respiratory tract (LRT) consists of:

👤 Larynx → Trachea → Bronchi → Lungs

▼ Its development begins during the 4th week of embryonic life and arises primarily from the ventral foregut.

1. 🌱 Early Development of the Respiratory System

● Laryngotracheal Groove

During the 4th week, a longitudinal depression called the laryngotracheal groove develops in the:

📍 Ventral wall of the caudal foregut

The groove is located in the primitive pharyngeal region.

Sequence:

Ventral foregut



Laryngotracheal groove



🫁 Respiratory diverticulum

🫁 Respiratory Diverticulum

The laryngotracheal groove evaginates outward to form the:

▼ Laryngotracheal diverticulum

✓ Also called the respiratory diverticulum or lung bud in its developing distal portion.

✓ Initially, it maintains an open communication with the foregut.

2. 🔄 Separation of Respiratory and Digestive Tracts

▼ As the respiratory diverticulum grows caudally, longitudinal folds called: Tracheoesophageal ridges develop on its lateral sides.

▼ These ridges fuse in the midline to form the: Tracheoesophageal septum

This separates the primitive foregut into:

Portion

Derivative

Ventral portion Laryngotracheal tube → respiratory

tract

Dorsal portion

Esophagus

3. 🧬 Germ-Layer Origins

💡 The respiratory tract develops from endoderm + splanchnic mesoderm.

● Endoderm → lining & glands

The endoderm forms:

✓ Epithelial lining of the respiratory tract

✓ Glands of the respiratory tract

✓ Laryngeal epithelium

✓ Tracheal epithelium

✓ Bronchial epithelium

✓ Alveolar epithelium

 Remember:

Endoderm = Epithelium + glands

● Splanchnic Mesoderm → supporting structures

The surrounding splanchnic mesoderm forms:

✓ Cartilage

✓ Smooth muscle

✓ Connective tissue

✓ Blood vessels

⚠ Important

Splanchnic mesoderm is NOT derived from neural crest cells.

So remember:

● Endoderm → epithelial lining + glands

● Splanchnic mesoderm → cartilage + smooth muscle + connective tissue

4. 🗣 Development of the Larynx

The larynx develops from the cranial-most part of the laryngotracheal tube.

🦴 Laryngeal Cartilages

Most laryngeal cartilages develop from mesenchyme associated with the: 4th and 6th pharyngeal arches

These include:

- ▼ Thyroid cartilage
- ▼ Cricoid cartilage
- ▼ Arytenoid cartilage
- ▼ Corniculate cartilage
- ▼ Cuneiform cartilage

 Epiglottis

The epiglottis develops from the hypopharyngeal eminence.

So:

 Laryngeal cartilages → 4th & 6th arches

 Epiglottis → hypopharyngeal eminence

5. Recanalization of the Larynx

Initially, proliferation of the laryngeal epithelium temporarily occludes the laryngeal lumen.

Later:

Epithelial proliferation



Temporary occlusion



Recanalization



Formation of:

- ✓ Vocal folds
- ✓ Vestibular folds

- ✓ Laryngeal ventricles

This recanalization occurs approximately during the 10th week.

Clinical correlation

Failure of recanalization → laryngeal atresia/stenosis

6. Development of the Trachea

The trachea develops from the portion of the laryngotracheal tube between the larynx and the site of bifurcation.

Derivatives

Endoderm →

- ▼ Respiratory epithelium
- ▼ Glands

Splanchnic mesoderm →

- ▼ Cartilage
- ▼ Smooth muscle
- ▼ Connective tissue

7. Development of Bronchi and Lungs

- ▼ The distal end of the respiratory diverticulum enlarges to form the: Lung bud
- ▼ The lung bud divides into two:

Primary bronchial buds

- ◆ Right primary bronchial bud
- ◆ Left primary bronchial bud

- ▼ These eventually form the right and left lungs.

▼ The growing lung buds extend into the pericardioperitoneal canals, which become associated with the developing pleural cavities.

Development of the Larynx

▼ The larynx develops from the cranial-most part of the laryngotracheal tube during the 4th week of embryonic development.

▼ The original communication between the developing respiratory tract and the primitive pharynx persists as the: Laryngeal inlet

This becomes the entrance through which air passes from the pharynx into the larynx.

1. Germ-Layer Origins

▼ The larynx has two major embryological components:

Endoderm → Epithelium + Glands

The endoderm of the laryngotracheal tube forms:

▼ Laryngeal epithelium

▼ Laryngeal glands

● Mesenchyme → Cartilage + Muscles + Connective Tissue

The laryngeal cartilages and muscles develop from mesenchyme associated with the:

4th and 6th pharyngeal arches

● Endoderm → Epithelium + glands

- 4th & 6th arches → laryngeal muscles + cartilages

2. Development of Laryngeal Cartilages

- ▼ Initially, the laryngeal opening is relatively slit-like.
- ▼ As the mesenchyme associated with the 4th and 6th pharyngeal arches proliferates around the opening:

Slit-like opening



Mesenchymal proliferation



T-shaped laryngeal opening

This mesenchyme subsequently forms the major laryngeal cartilages.

Major cartilages

-  Thyroid cartilage
-  Cricoid cartilage
-  Arytenoid cartilages
-  Corniculate cartilages
-  Cuneiform cartilages

 High-yield

4th + 6th pharyngeal arches → most laryngeal cartilages

3. Development of the Epiglottis

- ✓ The epiglottis is different from most other laryngeal cartilages.

- ✓ It develops from the caudal part of the hypopharyngeal eminence.
- ✓ The hypopharyngeal eminence develops in association with the 3rd and 4th pharyngeal arches.

Development

Hypopharyngeal eminence



 Epiglottis

The epiglottis enlarges considerably during fetal life and continues to grow after birth.

Epiglottis = hypopharyngeal eminence

4. Recanalization of the Larynx

▼ One of the most important events in laryngeal development is temporary epithelial occlusion followed by recanalization.

Step 1: Temporary Occlusion

The laryngeal epithelium proliferates rapidly.



The epithelial cells temporarily obliterate the laryngeal lumen.

So:

Open lumen → epithelial proliferation →  closed lumen

Step 2: Recanalization

By approximately the 10th week, the epithelial cells occupying the lumen break down.



The laryngeal lumen reopens.

 This process is called: Recanalization

5. Formation of the Vocal and Vestibular Folds

During recanalization, epithelial outgrowths form two pairs of folds.

Upper pair

Vestibular folds



False vocal cords

Lower pair

Vocal folds



True vocal cords

Between these folds develops the:

Laryngeal ventricle

The laryngeal ventricles are lateral recesses located between the:

Vestibular folds



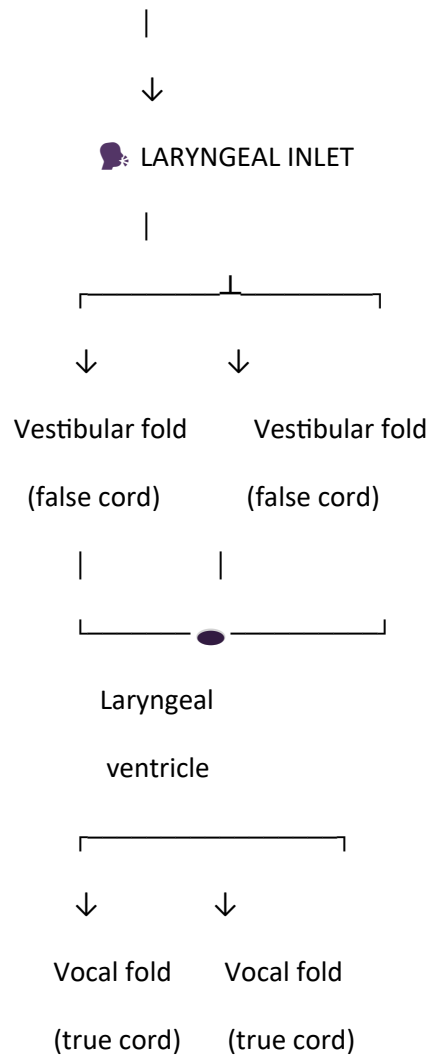
Laryngeal ventricles



Vocal folds

Visual Summary

PHARYNX



🚨 Clinical Correlation

Laryngeal Atresia

Cause: Failure of laryngeal recanalization

Normally:

Temporary occlusion → recanalization → open airway

But in laryngeal atresia:

Temporary occlusion → ❌ failure of recanalization



🚫 Upper fetal airway obstruction

🍷 Development of the Epiglottis

▼ The epiglottis is a specialized laryngeal structure that develops differently from most of the other laryngeal cartilages.

🌱 Embryological Origin

▼ The epiglottis develops from the:

Caudal part of the hypopharyngeal eminence

▼ The hypopharyngeal eminence is a swelling associated with the ventral ends of the 3rd and 4th pharyngeal arches.

Developmental sequence

3rd + 4th pharyngeal arches



Hypopharyngeal eminence



Caudal part enlarges



Epiglottis

The hypopharyngeal eminence is a mesenchymal swelling associated with the 3rd and 4th pharyngeal arches.

For exam purposes, the most important association is:

★ Epiglottis → hypopharyngeal eminence → 3rd & 4th pharyngeal arches

Position of the Developing Epiglottis

▼ The epiglottis develops near the:

- ✓ Root of the tongue
- ✓ Laryngeal inlet

As it grows, it becomes positioned behind the tongue and above the laryngeal inlet.

Its adult function is to help protect the airway during swallowing.

During swallowing

Food → pharynx



Epiglottis helps direct the bolus away from the laryngeal inlet



Esophagus

🧒 Postnatal Development

- ▼ The epiglottis continues to grow rapidly after birth.
- ▼ 📊 First 3 years
- ▼ It undergoes rapid growth and maturation, approaching its adult morphology by approximately 3 years of age.

Development of the Trachea

- ▼ The trachea develops during the 4th week from the laryngotracheal tube, which arises from the respiratory diverticulum.

- ▼ It forms the airway connecting the larynx to the main bronchi.

1. Embryological Origin

The trachea develops from the portion of the:

 Laryngotracheal tube between the larynx and its distal bifurcation

Developmental sequence

Ventral foregut



Laryngotracheal groove



Respiratory diverticulum



Laryngotracheal tube



 Trachea + main bronchi

2. Germ-Layer Origins

- ◆ The trachea has contributions from endoderm and splanchnic mesoderm.

● Endoderm → Inner lining

The endoderm of the laryngotracheal diverticulum forms:

✓ Respiratory epithelium

✓ Tracheal glands

 Remember

Endoderm → Epithelium + Glands

● Splanchnic Mesoderm → Supporting structures

The surrounding splanchnic mesoderm forms:

 Tracheal cartilage

 Smooth muscle

 Connective tissue

 Blood vessels

● Endoderm → epithelium + glands

● Splanchnic mesoderm → cartilage + smooth muscle + connective tissue

 3. Separation from the Esophagus

During the 4th week, the respiratory diverticulum develops from the ventral foregut.

As it grows caudally:

Tracheoesophageal ridges



fuse together



Tracheoesophageal septum

This separates the foregut into:

Portion	Derivative
Ventral	Laryngotracheal tube → respiratory tract
Dorsal	Esophagus

Ventral = Ventilation 🫁

Dorsal = Digestion 🍷

4. Differentiation of the Trachea

As development proceeds, the tracheal wall differentiates into its major components.

Around the 6th week

The developing tracheal lumen is lined by epithelium and surrounded by developing:

🦴 Cartilage

🧬 Connective tissue

💪 Smooth muscle

By approximately the 11th week

Further differentiation produces recognizable:

🦴 Cartilage

💪 Smooth muscle

💧 Glands

🧬 Connective tissue

✳️ The exact timing of individual histological maturation can vary, so the 6-week and 11-week landmarks are best treated as approximate developmental milestones, rather than absolute cutoffs.

🌿 5. Tracheal Bifurcation

✓ At its distal end, the developing respiratory tube divides into Two primary bronchial buds

▼ Right primary bronchial bud

▼ Left primary bronchial bud

These develop into the:

▼ Right main bronchus + right lung

▼ Left main bronchus + left lung

◆ Branching sequence

Primary bronchial buds



Secondary bronchi → lobar bronchi



Tertiary bronchi → segmental bronchi



Smaller airways



Bronchioles



Alveolar structures

6. Developmental Anomalies

A. Tracheoesophageal Fistula (TEF)

✓ An abnormal communication develops between:

Trachea ↔ Esophagus

✓ It results from abnormal partitioning of the foregut during early development.

Common clinical features

😬 Excessive salivation

 Respiratory distress

 Coughing/choking during feeding

 Cyanosis

 Aspiration

Classic association

Esophageal atresia + distal TEF

B. Tracheal Stenosis

- ▼ Stenosis = narrowing
- ▼ The tracheal lumen becomes abnormally narrow.
- ▼ Possible mechanisms include abnormal development or recanalization.

C. Tracheal Atresia

- ▼ Atresia = complete obstruction/absence of a patent lumen
- ▼ The tracheal airway is severely or completely blocked.
- ▼ This is much more severe than stenosis.

Development of the Bronchi and Lungs

▼ The bronchi and lungs develop from the distal part of the laryngotracheal (respiratory) diverticulum during the 4th–5th weeks of embryonic development.

- ▼ The distal respiratory diverticulum divides into two:

 Right bronchial bud + Left bronchial bud

These buds subsequently undergo repeated branching to establish the entire bronchial tree and respiratory portion of the lungs.

1. Formation of the Bronchial Buds

✓The distal end of the respiratory diverticulum enlarges and divides into Two primary bronchial buds.

➡ Right primary bronchial bud

➡ Left primary bronchial bud

✓These buds grow laterally into the: Pericardioperitoneal canals

✓These spaces contribute to the formation of the pleural cavities.

▼ Developmental sequence

Respiratory diverticulum



Right + left primary bronchial buds



Main bronchi



Lobar bronchi



Segmental bronchi



Smaller airways



Respiratory structures

2. Development of the Main Bronchi

 Beginning of the 5th week

✓ Each primary bronchial bud develops into a Principal (main) bronchus

Therefore:

 Right bronchial bud → right main bronchus

 Left bronchial bud → left main bronchus

Important anatomical relationship

The right main bronchus is:

 Shorter

↕ More vertical

 More in line with the trachea

The left main bronchus is:

 Longer

↔ More horizontal/transverse

 Clinical connection

Because the right main bronchus is more vertical, aspirated foreign bodies are more likely to enter the right bronchial tree.

3. Secondary (Lobar) Bronchi

✓ The main bronchi divide into:

✓ Secondary bronchi = lobar bronchi

✓ They correspond to the lobes of the lungs.

 Right lung

3 secondary bronchi

➔ Superior lobe

➔ Middle lobe

➔ Inferior lobe

 Left lung

2 secondary bronchi

➔ Superior lobe

➔ Inferior lobe

Right = 3 lobes = 3 lobar bronchi

Left = 2 lobes = 2 lobar bronchi

4. Tertiary (Segmental) Bronchi

✓Beginning around the 7th week, the secondary bronchi continue branching to form:

Tertiary bronchi = segmental bronchi

✓They supply the Bronchopulmonary segments

Approximate number:

Right lung → 10

Left lung → 8–10 depending on the classification used

△ Therefore, avoid treating 8 or 9 as an absolute number; anatomical descriptions vary because some segments may be fused.

5. Bronchopulmonary Segments

✓ Each segmental bronchus grows into a surrounding mass of mesenchyme.

✓ Segmental bronchus

Surrounding mesenchyme



Bronchopulmonary segment

✓ Each bronchopulmonary segment has its own:

✓ Segmental bronchus

✓ Branches of pulmonary artery

🎯 High-yield

Tertiary bronchus → bronchopulmonary segment

6. Continued Branching of the Bronchial Tree

▼ The bronchial tree continues branching extensively.

Sequence

Main bronchus



Lobar bronchi



Segmental bronchi



Smaller bronchi



Bronchioles



Terminal bronchioles



Respiratory bronchioles



Alveolar ducts



Terminal sacs



 Alveoli

✓ By approximately 24 weeks, many generations of branching have developed.

✓ The airway continues to mature after birth, and additional generations of branching develop postnatally.

7. Development of the Pleura

✓ As the lung buds expand into the pericardioperitoneal canals, they become surrounded by mesoderm.

Two layers of pleura develop:

Visceral Pleura

✓ Derived from:

▼ Splanchnic mesoderm

✓ It closely covers the developing lungs.

Parietal Pleura

✓ Derived from:

▼ Somatic mesoderm

✓ It lines the developing thoracic body wall.

● Pleural Cavity

✓ The space between the two pleural layers becomes the Pleural cavity

🔔 Notices

Visceral → organ → splanchnic

Parietal → body wall → somatic

8. Maturation of the Lungs

▼ Lung development occurs through four overlapping stages:

🌱 Pseudoglandular → 🩸 Canalicular → 🧫 Terminal Sac → 🫁 Alveolar

▼ The stages overlap because different regions of the lungs mature at slightly different rates.

1. Pseudoglandular Stage

🕒 5–17 weeks

The lungs look like an exocrine gland.

Major events:

- ▼ Extensive branching of the bronchial tree
- ▼ Formation of bronchi
- ▼ Formation of bronchioles
- ▼ Formation of terminal bronchioles

However:

- ✗ Respiratory bronchioles have not yet developed
- ✗ Functional alveolar gas exchange is not possible

Respiration

NOT possible

2. Canalicular Stage

 16–25 weeks

- ✓ The airway lumina become progressively larger.
- ✓ Each terminal bronchiole gives rise to Respiratory bronchioles, which give rise to Alveolar ducts

✓ At the same time:

 Pulmonary capillaries proliferate extensively.

- ✓ The developing respiratory epithelium becomes thinner and approaches the capillaries.

 Key event

Near the end of this stage:

Limited gas exchange becomes possible

because of Thin respiratory epithelium and Extensive capillary network

3. Terminal Sac Stage

 24 weeks → birth

- ▼ Numerous terminal sacs develop.
- ▼ These are the precursors of mature alveoli.
- ▼ The walls become progressively thinner.

 Type I Pneumocytes

These are:

- ▼ Thin squamous cells

Main function:

 Gas exchange

 Type II Pneumocytes

✓ These are secretory cells that produce:

- ▼ Pulmonary surfactant

✓ Surfactant is rich in phospholipids and Decreases surface tension

✓ Therefore:

↓ Alveolar collapse

↑ Lung compliance

Type I → Gas exchange 

Type II → Surfactant 

 Blood-Air Barrier

✓ During the terminal sac stage, the distance between Alveolar epithelium and Capillary endothelium becomes very small. This creates the Blood-air barrier.

- ▼ which allows efficient exchange of: $O_2 \leftrightarrow CO_2$

4. Alveolar Stage

 Late fetal period → childhood

- ▼ Terminal sacs mature into:

🌈 Alveolar sacs → mature alveoli

▼ A major proportion of alveolar development occurs after birth.

▼ The lungs increase in size primarily through:

★ Increase in the number of respiratory units and alveoli

rather than simply enlarging the existing alveoli.

Developmental Anomalies of the Lower Respiratory Tract

▼ Developmental anomalies of the lower respiratory tract (LRT) occur when normal embryological processes such as formation, separation, branching, recanalization, or maturation are disturbed.

▼ The major anomalies involve the larynx, trachea, bronchi, and lungs.

1. Laryngeal Atresia

✓ Laryngeal atresia is complete obstruction of the laryngeal lumen.

✓ Cause

It results from:

✗ Failure of laryngeal recanalization

Normally:

Epithelial proliferation



Temporary laryngeal occlusion



Recanalization



Patent laryngeal lumen

In laryngeal atresia:

Temporary occlusion → ❌ failure to recanalize



🚫 Upper fetal airway obstruction

🧠 Exam point

Failure of laryngeal recanalization → Laryngeal atresia

2. Tracheoesophageal Fistula (TEF)

✓ A tracheoesophageal fistula is an abnormal communication between: Trachea ↔ Esophagus

✓ Embryological cause

✓ During the 4th week, the respiratory diverticulum must separate from the esophagus through formation of the Tracheoesophageal septum

✓ Abnormal partitioning of the foregut can produce a persistent connection between the trachea and esophagus.

Abnormal foregut partitioning → TEF

★ TEF + Esophageal Atresia

✓ The most common clinically important pattern is:

◆ Esophageal atresia + distal TEF

✓ The esophagus ends blindly, while the distal esophageal segment communicates with the trachea.

Clinical consequences

📝 During feeding:

Milk → enters esophagus



Cannot pass normally into stomach



May enter respiratory tract through the fistula



Aspiration + respiratory distress

Possible findings:



Excessive salivation



Choking during feeding



Cyanosis



Respiratory distress



Aspiration pneumonia



TEF / Esophageal Atresia and Polyhydramnios

This is a high-yield clinical association.

Normally:

Amniotic fluid



Fetus swallows it



Passes through the GI tract



Fluid is absorbed

With esophageal atresia:

Fluid cannot pass normally through the esophagus into the stomach.



Amniotic fluid accumulates



Polyhydramnios

Esophageal atresia → ↓ fetal swallowing/absorption → ↑ amniotic fluid → Polyhydramnios

⚠ Important: Polyhydramnios is associated particularly with esophageal atresia, not simply with every TEF.

3. Tracheal Stenosis

Stenosis = narrowing

Therefore:

 Tracheal stenosis = abnormally narrowed tracheal lumen

Clinical effect

↓ Airflow



 Respiratory difficulty

The severity depends on the degree and location of narrowing.

4. Tracheal Atresia

Atresia = complete obstruction or failure of formation of a patent lumen.

Therefore:

 Tracheal atresia → severe/complete airway obstruction

It is considerably more severe than stenosis.

5. Pulmonary Agenesis

Agenesis = complete failure of development

Therefore Pulmonary agenesis = complete absence of a lung due to failure of development of the corresponding lung bud.

It can be:

- ▼ Unilateral
- ▼ Rarely bilateral

Exam point

Agenesis = organ never develops

6. Pulmonary Hypoplasia

✦ Hypoplasia = incomplete development

- ▼ Unlike agenesis, the lung is present, but it is inadequately developed.

Pulmonary hypoplasia

- ▼ ↓ Lung size
- ▼ ↓ Number/size of airways
- ▼ ↓ Alveoli
- ▼ ↓ Respiratory capacity

★ Important association

Oligohydramnios → pulmonary hypoplasia

This can occur particularly with severe fetal renal abnormalities because reduced fetal urine production causes reduced amniotic fluid.

Distinguish

Agenesis = absent ✖

Hypoplasia = underdeveloped ●

7. Congenital Lung Cysts

✓ Abnormal development of the bronchial tree can result in Congenital cystic lesions

✓ These may appear as abnormal cystic spaces within the developing lung.

✓ They can interfere with:

- ▼ Normal lung development

- ▼ Airway function

- ▼ Respiratory function

8. Accessory Lung

✓ An accessory lung refers to additional lung tissue separate from the normal pulmonary structures.

✓ It may have an abnormal or separate bronchial connection.

Key distinction

Accessory lung = extra lung tissue

✓ It should be distinguished from other congenital pulmonary malformations in which abnormal lung tissue has a different vascular or airway arrangement.